ClinValAI: A framework for developing Cloud-based infrastructures for the External Clinical Validation of AI in Medical Imaging

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INTRODUCTION

- Generalizability challenges and latent biases necessitate external clinical validation of AI algorithms.
- Patient privacy concerns, intellectual property rights, and diverse model requirements impede validation efforts.
- ClinValAI addresses the limited technical guidance and barriers associated with the clinical validation of AI.
- Our framework can enhance a health institution's decision-making process for identifying the most suitable AI model for their specific application, to improve the health of their target population.

METHODS

- ClinValAI features dedicated and easily-customizable workflows to facilitate external validation of AI algorithms.

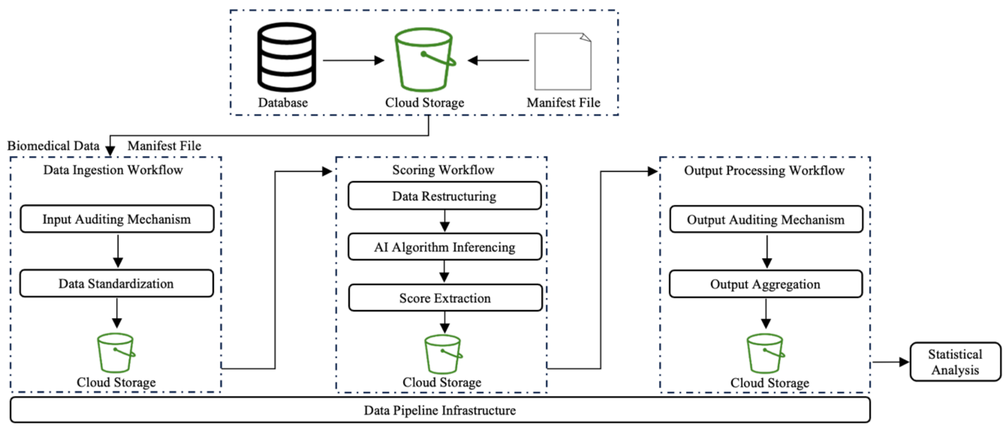

Fig. 1. Conceptual overview of the ClinValAI framework.

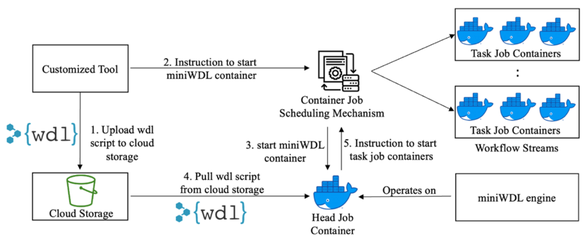

Fig. 2. Job scheduling and batch processing orchestration

- ClinValAI features multiple pipelines for data processing and AI validation through a combination of series and parallel jobs.
- A batch of data is processed serially by leveraging numerous Docker containers running sequentially, and each batch is processed independently of other batches in a parallel fashion.
- ClinValAI's data pipeline enables leveraging the full potential of the cloud computing environment.

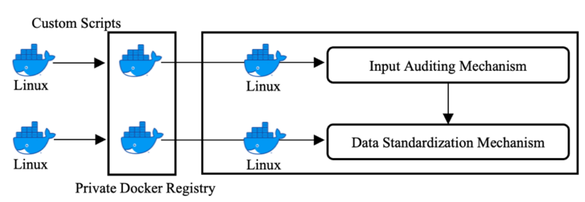

Fig. 3. Data ingestion workflow

- The input auditing mechanism verifies if the data can be processed and are aligned with the AI model's prerequisites.
- The data standardization mechanism facilitates information imputation, data preprocessing, and ensures data quality.

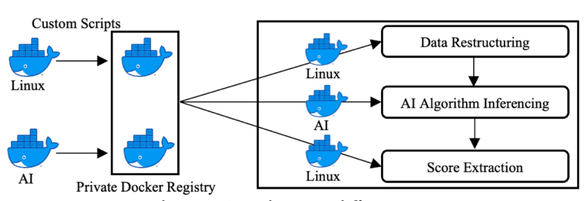

Fig. 4. Scoring workflow

- Data restructuring helps cater to model-specific requirements.
- Effective customization of models' Docker images facilitates robust external validation of AI algorithms.
- ClinValAI enables retrieving specific outputs of interest, such as image-, exam-, or patient-level scores.

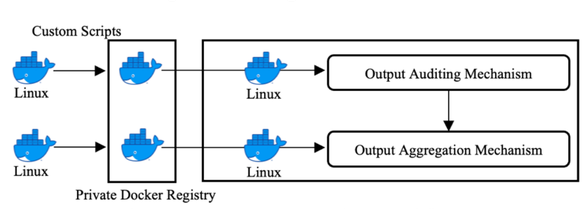

Fig. 5. Output processing workflow

- The output auditing mechanism verifies if results have been produced for all inputs and if the generated files comply with the algorithm's expected outputs.
- ClinValAI identifies samples with missing output data, irretrievable scores, and corrupted output files to enable analysis of samples to be re-scored.

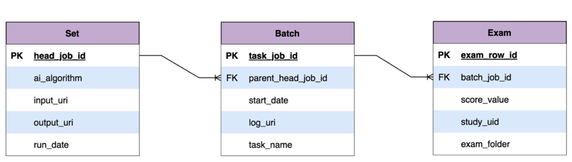

Fig. 6. ERD for the aggregated output data

- Analyzing the outputs from individual workflows in the distributed format of cloud storage can be cumbersome.
- ClinValAI systematically aggregates details by appending all results to a relational database. During statistical analysis, this database is pulled to analyze findings.

EVALUATION AND RESULTS

- To evaluate the utility of our framework, we used our ClinValAI-based cloud infrastructure to perform a rigorous external validation of Mirai, a state-of-the-art open-source mammography-based AI algorithm for breast cancer risk prediction.
- Mirai scores were generated for 26,449 screening exams from 14,291 patients from four imaging facilities in the University of Washington Medicine health system.
- To help distinguish between breast cancer detection vs. risk prediction performance, we performed the analyses using all available exams and then repeated the analyses after excluding exams that had a breast cancer diagnosis within 6 months.

Table 1. Discrimination performance of Mirai in the University of Washington and 7 previously reported cohorts.

	1-Year AUC (95% CI)	5-Year AUC (95% CI)	C-index (95% CI)
All Exams			
University of Washington, USA	**0.81 (0.75-0.86)**	**0.70 (0.67-0.72)**	**0.70 (0.67-0.72)**
MGH, USA	0.84 (0.80-0.87)	0.76 (0.73-0.79)	0.75 (0.72-0.78)
Novant, USA	0.78 (0.73-0.84)	0.75 (0.70-0.80)	0.75 (0.70-0.80)
Emory, USA	0.83 (0.81-0.86)	0.76 (0.74-0.79)	0.77 (0.75-0.79)
Maccabi-Assuta, Israel	0.86 (0.81-0.91)	0.75 (0.71-0.79)	0.77 (0.73-0.81)
Karolinska, Sweden	0.90 (0.89-0.92)	0.78 (0.76-0.80)	0.81 (0.79-0.82)
CGMH, Taiwan	0.90 (0.87-0.93)	0.79 (0.75-0.82)	0.79 (0.76-0.83)
Barretos, Brazil	0.89 (0.86-0.93)	0.82 (0.78-0.86)	0.84 (0.81-0.88)
Excluding Cancers within 6 Months			
University of Washington, USA	**0.72 (0.56-0.84)**	**0.68 (0.65-0.71)**	**0.68 (0.65-0.70)**
MGH, USA	0.71 (0.60-0.84)	0.71 (0.68-0.75)	0.69 (0.66-0.73)
Novant, USA	N/A	0.72 (0.66-0.79)	0.72 (0.66-0.79)
Emory, USA	0.74 (0.66-0.84)	0.71 (0.68-0.74)	0.69 (0.66-0.72)
Maccabi-Assuta, Israel	N/A	0.68 (0.62-0.74)	0.70 (0.64-0.76)
Karolinska, Sweden	N/A	0.71 (0.69-0.73)	0.71 (0.69-0.74)
CGMH, Taiwan	0.84 (0.72-0.99)	0.70 (0.66-0.75)	0.70 (0.66-0.75)
Barretos, Brazil	0.87 (0.80-0.94)	0.75 (0.70-0.80)	0.78 (0.74-0.83)

- Mirai's performance on the UW cohort was similar to previously reported results in other cohorts, though they were still on the lower end of the range.

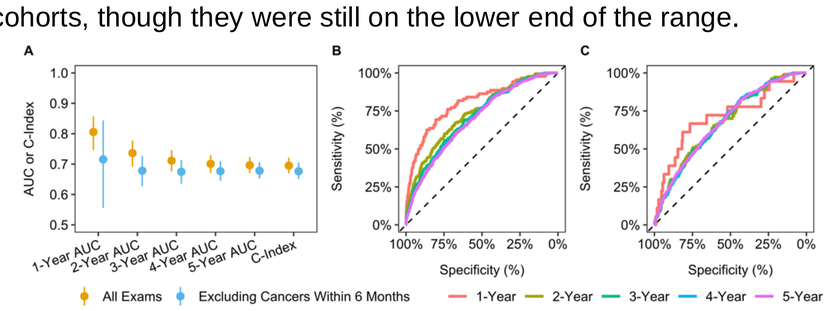

Fig. 7. Discrimination performance for years 1-5 following the index examination

- There were no statistically significant differences in the c-index between subgroups defined by age, race, and breast density.

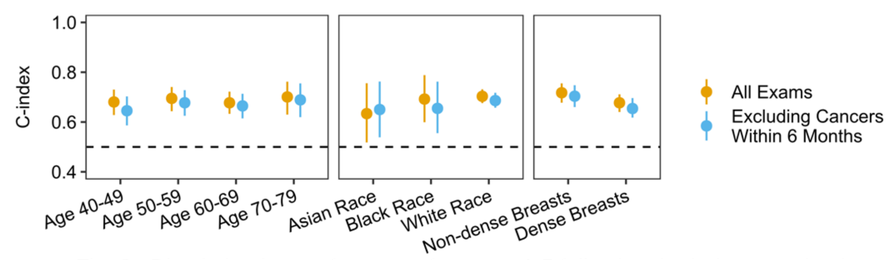

Fig. 8. Discrimination performance for years 1-5 following the index examination

- When all exams were included, Mirai significantly overestimated risk in years 1 and 2, but was overall reasonably well calibrated for the later years. When exams with diagnoses within 6 months were excluded, the metrics substantially worsened.

Table 2. Calibration plots with predicted risks grouped into deciles.

	All Exams				After Excluding Cancers within 6 Months			
	Overall Calibration*		Calibration Slope†		Overall Calibration*		Calibration Slope†	
Timeframe	Estimate (%)	(95% CI)	Estimate	(95% CI)	Estimate (%)	(95% CI)	Estimate	(95% CI)
1-year risk	-0.10	(-0.17, -0.03)	0.83	(0.52, 1.22)	-0.36	(-0.39, -0.32)	0.05	(0.00, 0.12)
2-year risk	-0.15	(-0.25, -0.04)	0.76	(0.51, 1.06)	-0.40	(-0.48, -0.32)	0.21	(0.08, 0.36)
3-year risk	-0.13	(-0.29, 0.03)	0.89	(0.60, 1.20)	-0.38	(-0.53, -0.26)	0.40	(0.20, 0.63)
4-year risk	-0.05	(-0.25, 0.14)	0.90	(0.63, 1.20)	-0.31	(-0.49, -0.12)	0.50	(0.27, 0.76)
5-year risk	0.09	(-0.15, 0.33)	1.03	(0.75, 1.32)	-0.16	(-0.38, 0.07)	0.66	(0.44, 0.91)

DISCUSSION

- AI algorithms have demonstrated promising results in transforming clinical medicine. Concerns regarding the under-performance of over-parametrized deep learning models on out-of-distribution validation sets can lead to reservations about the generalizability of AI algorithms.
- Our work addresses critical challenges associated with external validation efforts and provides an easily customizable and cloud-agnostic framework to build scalable infrastructures to audit and monitor AI algorithms in clinical medicine.
- ClinValAI promotes the external validation of AI algorithms prior to their integration into clinical workflows to reliably understand their real-world performance in healthcare settings and their impact on patient care, health, and safety.
- Our framework can be leveraged to evaluate the generalizability of deep learning models on healthcare data from diverse demographics to analyze the differences in performance across various sub-populations and identify latent biases in AI algorithms.
- ClinValAI is equipped to continuously monitor model performance vis-à-vis data drift and model drift, and supports a streamlined feedback mechanism with vendors, encouraging targeted algorithm fine-tuning efforts and fostering enhanced academic-industry partnerships.
- Developers can analyze performance based on detailed findings from the results communicated by ClinValAI. This encourages the development of explainable AI models to better reason performance, thereby enhancing the potential of receiving clinicians' trust.
- Our framework can facilitate the detection of AI algorithms' failure modes, understand its potential to function as a standalone tool for diagnostic applications, and study its capabilities in reducing the burden on the medical fraternity.
- By enabling large-scale external validation efforts on data from diverse cohorts, our work has the potential to overcome health disparities and foster health equity by promoting the development of robust, interpretable, and generalizable algorithms for healthcare applications.

ACKNOWLEDGEMENTS

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